SE Lab 6: Code and Branch Coverage

Name: Venkata Srivaibhav Nukaraju

SRN: PES2UG24CS579

Section: 5 J

Course: UE24CS341A Software Engineering

1. Links

Assigned repository: https://github.com/SETAPESU26/L6_L7_19_parking_fee

Module: parking_fee.py. The module computes parking fees, assigns slots and prices monthly passes.

Test files: CodeCoverageTest_PES2UG24CS579_parking_fee.py and BranchCoverageTest_PES2UG24CS579_parking_fee.py.

Two commits add the test files and the screenshots, one commit per coverage stage.

2. Coverage Results

Test stage	Tests	Line coverage	Branch coverage	Missing
Starter tests	2	44%	40%	6, 8, 12-15, 17, 19, 25, 27, 29, 34-42
Code coverage tests	11	100%	98%	13->15
Branch coverage tests	18	100%	100%	None

The figures apply to parking_fee.py, which holds 34 statements and 24 branches.

3. Analysis

The starter tests reach only the free-stay path and the default slot. The code coverage tests add one test for each missed line, including both ValueError paths. One test with 90 minutes, an EV and a lost ticket runs lines 12 to 19.

Line coverage reaches 100%, but branch coverage stays at 98%. Branch 13->15 shows that the condition minutes % 60 != 0 never evaluates to False. Every charged test uses a partial hour, so the exact-hour path never runs. The 120-minute test closes this gap.

The branch tests also check the 30 and 31 minute boundary and the EV truck case. Another test checks a full lot with a disabled driver. Coverage.py counts a compound condition as one branch, so the EV truck test does not change the number. It still confirms that a truck gets no EV discount.

4. Output Screenshots

The screenshots are rendered headless from the real command output, so the window focus stays unchanged.

Screenshot 1: Line coverage with the code coverage tests

```
PES2UG24CS579@Venkata-Srivaibhav-Nukaraju:~/L6_L7_19_parking_fee/Lab 6$ coverage run -m pytest CodeCoverageTest_PES2UG24CS579_parking_fee.py
============================= test session starts ==============================
platform linux -- Python 3.14.8, pytest-9.2.0, pluggy-1.7.0
rootdir: /HDD/lab/College/Year-3/Sem-5/SE/Lab6/L6_L7_19_parking_fee/Lab 6
collected 11 items

CodeCoverageTest_PES2UG24CS579_parking_fee.py ...........               [100%]

============================== 11 passed in 0.02s ==============================
PES2UG24CS579@Venkata-Srivaibhav-Nukaraju:~/L6_L7_19_parking_fee/Lab 6$ coverage report -m
Name                                              Stmts   Miss  Cover   Missing
-------------------------------------------------------------------------------
CodeCoverageTest_PES2UG24CS579_parking_fee.py      27      0   100%
parking_fee.py                                     34      0   100%
-------------------------------------------------------------------------------
TOTAL                                              61      0   100%
```

Screenshot 2: Branch coverage with the code coverage tests

```
PES2UG24CS579@Venkata-Srivaibhav-Nukaraju:~/L6_L7_19_parking_fee/Lab 6$ coverage run --branch -m pytest CodeCoverageTest_PES2UG24CS579_parking_fee.py
============================= test session starts ==============================
platform linux -- Python 3.14.8, pytest-9.2.0, pluggy-1.7.0
rootdir: /HDD/lab/College/Year-3/Sem-5/SE/Lab6/L6_L7_19_parking_fee/Lab 6
collected 11 items

CodeCoverageTest_PES2UG24CS579_parking_fee.py ...........               [100%]

============================== 11 passed in 0.02s ==============================
PES2UG24CS579@Venkata-Srivaibhav-Nukaraju:~/L6_L7_19_parking_fee/Lab 6$ coverage report -m
Name                                              Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------------------
CodeCoverageTest_PES2UG24CS579_parking_fee.py      27      0      0      0   100%
parking_fee.py                                     34      0     24      1    98%   13->15
-----------------------------------------------------------------------------------------------
TOTAL                                              61      0     24      1    99%
```

Screenshot 3: Line coverage with the branch coverage tests

```
PES2UG24CS579@Venkata-Srivaibhav-Nukaraju:~/L6_L7_19_parking_fee/Lab 6$ coverage run -m pytest BranchCoverageTest_PES2UG24CS579_parking_fee.py
============================= test session starts ==============================
platform linux -- Python 3.14.8, pytest-9.2.0, pluggy-1.7.0
rootdir: /HDD/lab/College/Year-3/Sem-5/SE/Lab6/L6_L7_19_parking_fee/Lab 6
collected 18 items

BranchCoverageTest_PES2UG24CS579_parking_fee.py ..................      [100%]

============================== 18 passed in 0.03s ==============================
PES2UG24CS579@Venkata-Srivaibhav-Nukaraju:~/L6_L7_19_parking_fee/Lab 6$ coverage report -m
Name                                              Stmts   Miss  Cover   Missing
-------------------------------------------------------------------------------
BranchCoverageTest_PES2UG24CS579_parking_fee.py      41      0   100%
parking_fee.py                                     34      0   100%
-------------------------------------------------------------------------------
TOTAL                                              75      0   100%
```

Screenshot 4: Branch coverage with the branch coverage tests

```
PES2UG24CS579@Venkata-Srivaibhav-Nukaraju:~/L6_L7_19_parking_fee/Lab 6$ coverage run --branch -m pytest BranchCoverageTest_PES2UG24CS579_parking_fee.py
============================= test session starts ==============================
platform linux -- Python 3.14.8, pytest-9.2.0, pluggy-1.7.0
rootdir: /HDD/lab/College/Year-3/Sem-5/SE/Lab6/L6_L7_19_parking_fee/Lab 6
collected 18 items

BranchCoverageTest_PES2UG24CS579_parking_fee.py ..................       [100%]

============================== 18 passed in 0.03s ==============================
PES2UG24CS579@Venkata-Srivaibhav-Nukaraju:~/L6_L7_19_parking_fee/Lab 6$ coverage report -m
Name                                            Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------------------------------
BranchCoverageTest_PES2UG24CS579_parking_fee.py      41      0      0      0   100%
parking_fee.py                                       34      0     24      0   100%
-------------------------------------------------------------------------------------------
TOTAL                                                75      0     24      0   100%
```

Screenshot 5: Git log with the two coverage commits

```
PES2UG24CS579@Venkata-Srivaibhav-Nukaraju:~/L6_L7_19_parking_fee/Lab 6$ git log --oneline
6bad05c Add branch coverage tests: 100% branch coverage on parking_fee.py
6d3fa9d Add code coverage tests: 100% line coverage on parking_fee.py
af37455 Add Lab 6 parking fee
```